

Logical volumes: Logical volumes are created from free physical extents in volume group, and provide the storage device that can be used by users and operating systems. Logical volumes are a collection of logical extents that map to physical extents (a physical extent is the smallest storage chunk of a physical volume). By default, each logical extent maps to one PE. But this mapping can be changed by using specific options.

Implementing LVM storage

Preparing the physical device

We have to create a partition of any size but remember to change the type of partition to the Linux LVM by using code 8e.

Creating a physical volume

Physical volume can be created by the `pvcreate` command as shown in Figure 7. `pvdisplay` is used to verify the created physical volume.

Creating a volume group

A volume group can be created by `vgcreate` and then by providing the name of the volume group. `vgdisplay` is used to verify volume group.

Creating a logical volume

An LVM can be created by using the `lvcreate` command, whereby we specify the size of the LVM using the `-L` option and the name of the LVM with the `-n` option, after which the volume group's name needs to be provided.

The logical volume size can be extended, provided space is available in the associated volume group. If so, then the `lvresize` command can be used to extend the LVM. You can verify the

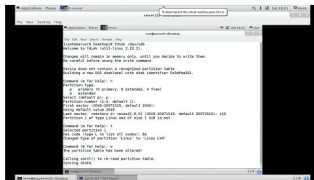


Figure 6: The creation of a physical disk `/dev/vdb1` of size 1GiB



Figure 7: The creation of a physical volume



Figure 8: The creation of a volume group named `myvol`



Figure 9: The creation of an LVM (of size 200MiB), named `mylvm` from the volume group `myvol`

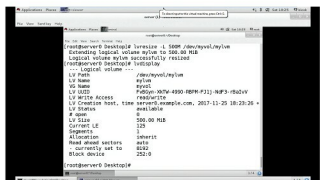


Figure 10: Logical volume size has been extended to 500MiB

size of the extended LVM by using the `lvdisplay` command.

To conclude, here are some reasons why disk partitioning is necessary or beneficial:

- Allows multi-booting of different operating systems from the same disk.
- Separates the operating system and program files from the user files.
- Creates separate areas for OS virtual memory swapping.
- Limits disk usage to improve performance of diagnostic tools and backup imaging. 

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